# Signals and Systems

## Lecture 11: Continuous-time Fourier Series

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Zhejiang University

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Adapted from the materials provided on
the MIT OpenCourseWare

# Fourier Representations

Fourier series represent **signals** in terms of **sinusoids**.

$\rightarrow$ leads to a new representation for **systems** as **filters**.

# Fourier Series: Review

Determining harmonic components of a periodic signal.

$$a_k = \frac{1}{T}\int_T x(t)e^{-j\frac{2\pi}{T}kt}dt \qquad \text{(“analysis” equation)}$$

$$x(t) = x(t+T) = \sum_{k=-\infty}^{\infty} a_k e^{j\frac{2\pi}{T}kt} \qquad \text{(“synthesis” equation)}$$

# Simplifying Math By Using Complex Numbers

Our biggest simplification comes from **Euler's formula**, which relates complex exponentials to trigonometric functions (Leonhard Euler, 1748).

$$e^{j\theta} = \cos\theta + j\sin\theta$$

where $j = \sqrt{-1}$.

Richard Feynman called this "the most remarkable formula in mathematics."

# Difference: Negative k

The complex exponential form of the series has positive and negative $k$'s.

$$f(t) = \sum_{k=-\infty}^{\infty} a_k e^{jk\omega_o t}$$

Only positive values of $k$ are used in the trig form.

$$f(t) = c_0 + \sum_{k=1}^{\infty} c_k \cos(k\omega_o t) + \sum_{k=1}^{\infty} d_k \sin(k\omega_o t)$$

Q: Why? What does negative k mean?

The negative $k$'s are required by Euler's formula.

$$e^{jk\omega_o t} = \cos(k\omega_o t) + j \sin(k\omega_o t)$$

$$\cos(k\omega_o t) = \mathrm{Re}\{e^{jk\omega_o t}\} = \frac{1}{2}\left(e^{jk\omega_o t} + e^{-jk\omega_o t}\right)$$

$$\sin(k\omega_o t) = \mathrm{Im}\{e^{jk\omega_o t}\} = \frac{1}{2j}\left(e^{jk\omega_o t} - e^{-jk\omega_o t}\right)$$

The negative $k$ do not indicate negative frequencies. They are the mathematical result of representing sinusoids with complex exponentials.

# CTFS: Complex Exponential Form

Represent periodic signal f(t) as a sum of harmonically-related *complex exponentials:*

$$f(t) = f(t+T) = \sum_{k=-\infty}^{\infty} a_k e^{j\omega_o k t}$$

Q: How to find coefficients?

We can "sift" out the component at $l\omega_o$ by multiplying both sides by $e^{-jl\omega_o t}$ and integrating over a period.

Let's try it!

$$e^{j2\pi} = 1; e^{j\pi} = -1; e^{j\pi/2} = j;$$

$$\int_T f(t)e^{-j\omega_o l t}dt = \int_T \sum_{k=-\infty}^{\infty} a_k e^{j\omega_o k t} e^{-j\omega_o l t}dt = \sum_{k=-\infty}^{\infty} a_k \int_T e^{j\omega_o (k-l)t}dt = \begin{cases} Ta_l & \text{if } l = k \\ 0 & \text{otherwise} \end{cases}$$

Solving for $a_l$ provides an explicit formula for the coefficients:

$$a_k = \frac{1}{T}\int_T f(t)e^{-j\omega_o k t}dt\,; \quad \text{where } \omega_o = \frac{2\pi}{T}\,.$$

Q: why can we do this?

# Fourier Series: Orthogonal Decomposition

Determining harmonic components of a periodic signal.

$$a_k = \frac{1}{T}\int_T x(t)e^{-j\frac{2\pi}{T}kt}dt \qquad \text{("analysis" equation)}$$

$$x(t) = x(t+T) = \sum_{k=-\infty}^{\infty} a_k e^{j\frac{2\pi}{T}kt} \qquad \text{("synthesis" equation)}$$

We can think of Fourier series as an **orthogonal decomposition**.

# Orthogonal Decompositions

**Vector representation of 3-space:** let $\bar{r}$ represent a vector with components $\{x,\ y,\ \text{and}\ z\}$ in the $\{\hat{x},\ \hat{y},\ \text{and}\ \hat{z}\}$ directions, respectively.

$$x = \bar{r} \cdot \hat{x}$$
$$y = \bar{r} \cdot \hat{y} \qquad \text{(“analysis” equations)}$$
$$z = \bar{r} \cdot \hat{z}$$

$$\bar{r} = x\hat{x} + y\hat{y} + z\hat{z} \qquad \text{(“synthesis” equation)}$$

# Orthogonal Decompositions

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**Fourier series:** let $x(t)$ represent a signal with harmonic component $\{a_0,\ a_1,\ \ldots,\ a_k\}$ for harmonics $\{e^{j0t},\ e^{j\frac{2\pi}{T}t},\ \ldots,\ e^{j\frac{2\pi}{T}kt}\}$ respectively.

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# Orthogonal Decompositions

Integrating over a period **sifts** out the $k^{\text{th}}$ component of the series.

Sifting as a dot product:

$$x = \bar{r} \cdot \hat{x} \equiv |\bar{r}||\hat{x}| \cos\theta$$

Sifting as an inner product:

$$a_k = e^{j\frac{2\pi}{T}kt} \cdot x(t) \equiv \frac{1}{T}\int_T x(t)e^{-j\frac{2\pi}{T}kt}dt$$

where

$$a(t) \cdot b(t) = \frac{1}{T}\int_T a^*(t)b(t)dt\,.$$

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where

$$a(t) \cdot b(t) = \frac{1}{T}\int_T a^*(t)b(t)dt\,.$$

The complex conjugate ($^*$) makes the inner product of the $k^{\text{th}}$ and $m^{\text{th}}$ components equal to 1 iff $k = m$:

$$\frac{1}{T}\int_T \left(e^{j\frac{2\pi}{T}kt}\right)^* \left(e^{j\frac{2\pi}{T}mt}\right)dt = \frac{1}{T}\int_T e^{-j\frac{2\pi}{T}kt}e^{j\frac{2\pi}{T}mt}dt = \begin{cases} 1 & \text{if } k = m \\ 0 & \text{otherwise} \end{cases}$$

# Check Yourself

How many of the following pairs of functions are orthogonal ($\perp$) in $T = 3$?

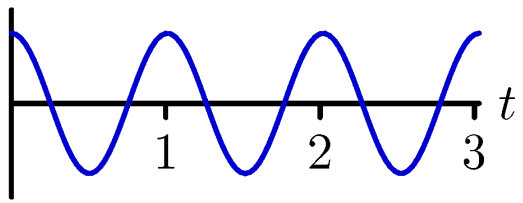
1. $\cos 2\pi t \perp \sin 2\pi t$ ?
2. $\cos 2\pi t \perp \cos 4\pi t$ ?
3. $\cos 2\pi t \perp \sin \pi t$ ?
4. $\cos 2\pi t \perp e^{j2\pi t}$ ?

# Check Yourself

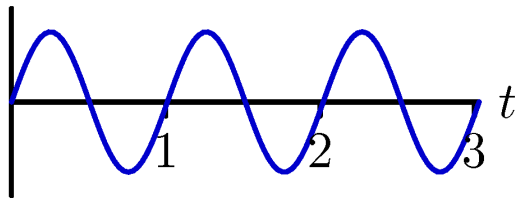
How many of the following are orthogonal ($\perp$) in $T = 3$?

$\cos 2\pi t \perp \sin 2\pi t$ ?

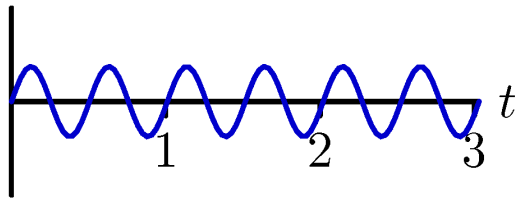
$\cos 2\pi t$




$\sin 2\pi t$




$\cos 2\pi t \sin 2\pi t$



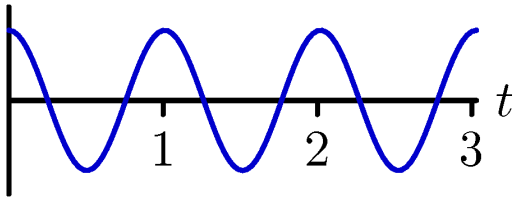

$\int_0^3 dt = 0$ therefore YES

# Check Yourself

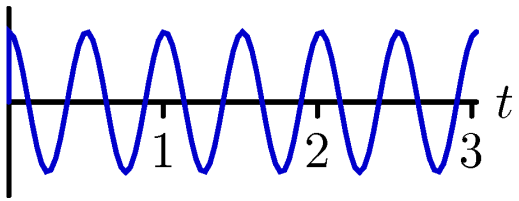
How many of the following are orthogonal ($\perp$) in $T = 3$?

$\cos 2\pi t \perp \cos 4\pi t$ ?

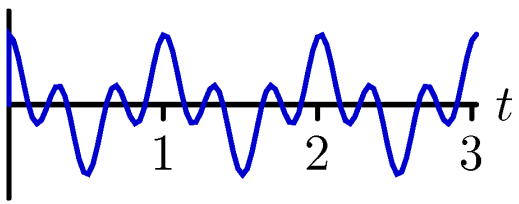
$\cos 2\pi t$




$\cos 4\pi t$




$\cos 2\pi t \cos 4\pi t$

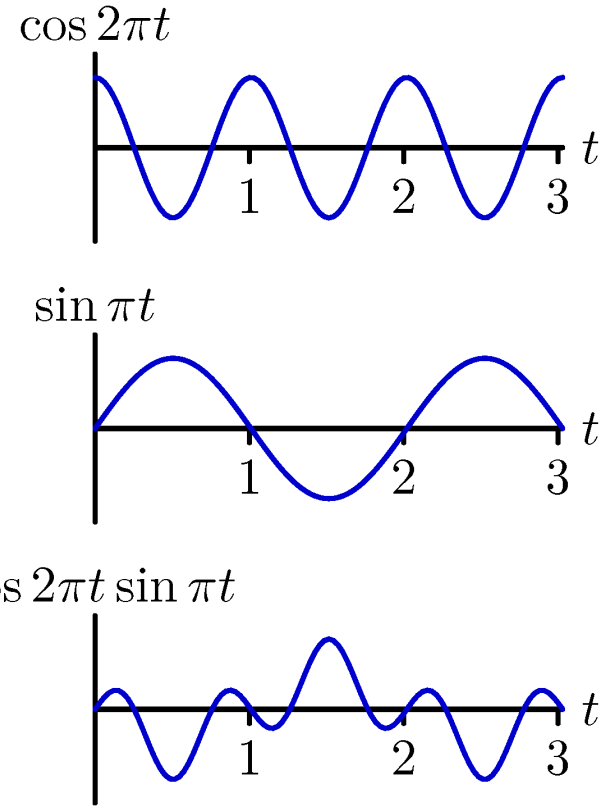



$\int_0^3 dt = 0$ therefore YES

# Check Yourself

How many of the following are orthogonal ($\perp$) in $T = 3$?

$\cos 2\pi t \perp \sin \pi t$ ?




$\int_0^3 dt \neq 0$ therefore NO

# Check Yourself

How many of the following are orthogonal ($\perp$) in $T = 3$?

$\cos 2\pi t \perp e^{2\pi t}$ ?

$$e^{2\pi t} = \cos 2\pi t + j \sin 2\pi t$$

$\cos 2\pi t \perp \sin 2\pi t$ but not $\cos 2\pi t$

Therefore NO

# Check Yourself

How many of the following pairs of functions are orthogonal ($\perp$) in $T = 3$? 2

1. $\cos 2\pi t \perp \sin 2\pi t$ ? √
2. $\cos 2\pi t \perp \cos 4\pi t$ ? √
3. $\cos 2\pi t \perp \sin \pi t$ ? ×
4. $\cos 2\pi t \perp e^{j2\pi t}$ ? ×

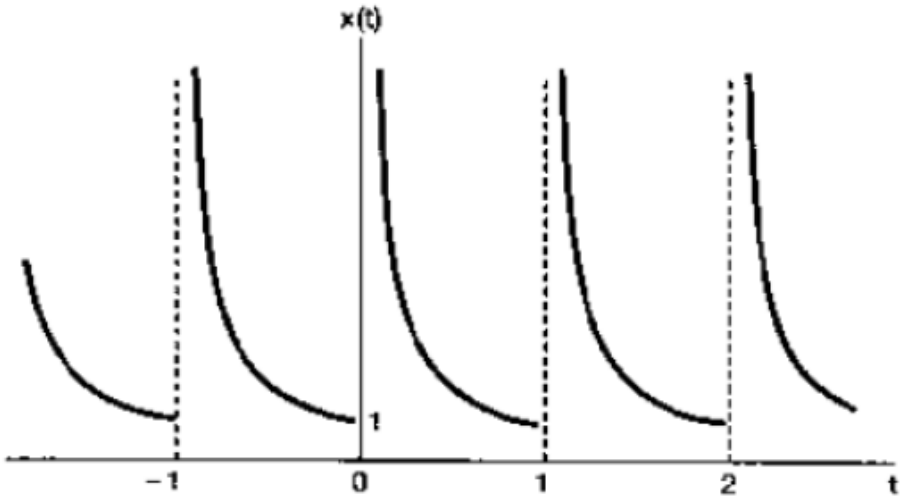
# Convergence: The Dirichlet Conditions

- **Condition 1:**

  Over any period, x(t) must be absolutely integrable

$$\int_T |x(t)|dt < \infty$$

**Counter example**




$$x(t) = \frac{1}{t}, \quad 0 < t \le 1$$

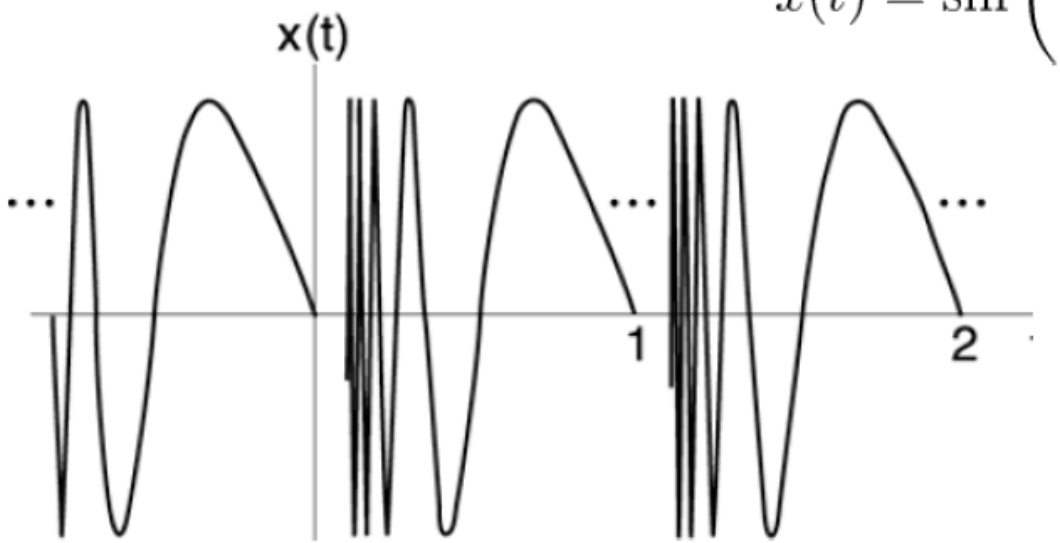
# Convergence: The Dirichlet Conditions

- **Condition 2:**

  In any finite interval of time, x(t) is of bounded variation; i.e. there are no more than a finite number of maxima and minima during any single period of the signal.

**Counter example**

$$x(t) = \sin\left(\frac{2\pi}{t}\right) \quad 0 < t \leq 1$$

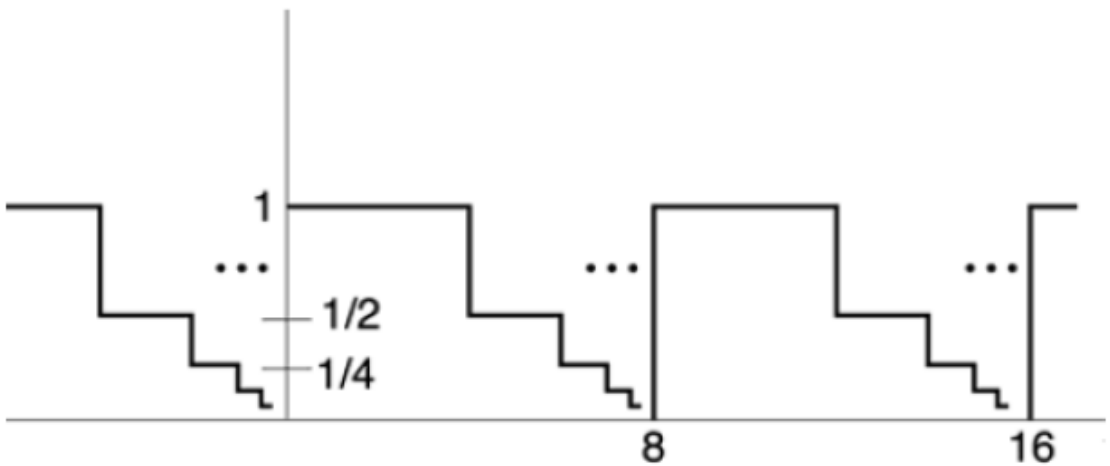


# Convergence: The Dirichlet Conditions

- **Condition 3:**

  In any finite interval of time, there are only a finite number of discontinuities. Each of these discontinuities is finite.

**Counter example**

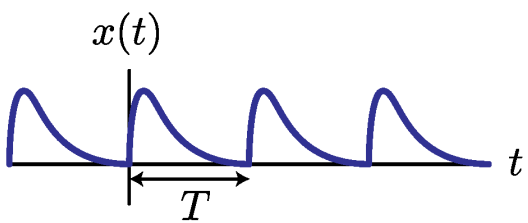


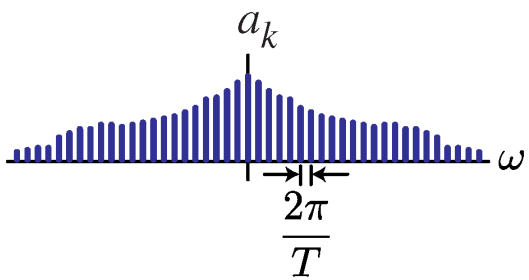
# Review: Fourier Series

Representing periodic signals as sums of **sinusoids**.

$$a_k = \frac{1}{T}\int_T x(t)e^{-j\frac{2\pi}{T}kt}dt \qquad \text{("analysis" equation)}$$

$$x(t) = x(t+T) = \sum_{k=-\infty}^{\infty} a_k e^{j\frac{2\pi}{T}kt} \qquad \text{("synthesis" equation)}$$



# Filtering

Notion of a filter.

LTI systems

- cannot create new frequencies.
- can scale magnitudes and shift phases of existing components.

# Filtering

The output of an LTI system is a "filtered" version of the input.

Input: Fourier series $\rightarrow$ sum of complex exponentials.

$$x(t) = x(t+T) = \sum_{k=-\infty}^{\infty} a_k e^{j\frac{2\pi}{T}kt}$$

Complex exponentials: eigenfunctions of LTI systems.

$$e^{j\frac{2\pi}{T}kt} \rightarrow H(j\frac{2\pi}{T}k)e^{j\frac{2\pi}{T}kt}$$

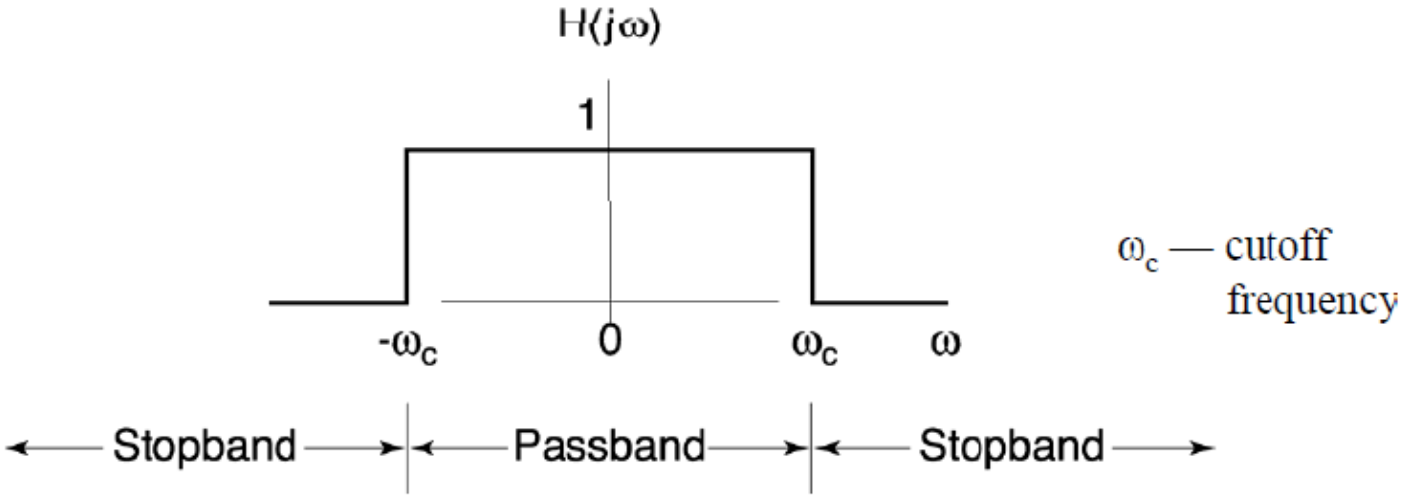
Output: same eigenfunctions, amplitudes/phases set by system.

$$x(t) = \sum_{k=-\infty}^{\infty} a_k e^{j\frac{2\pi}{T}kt} \rightarrow y(t) = \sum_{k=-\infty}^{\infty} a_k H(j\frac{2\pi}{T}k)e^{j\frac{2\pi}{T}kt}$$

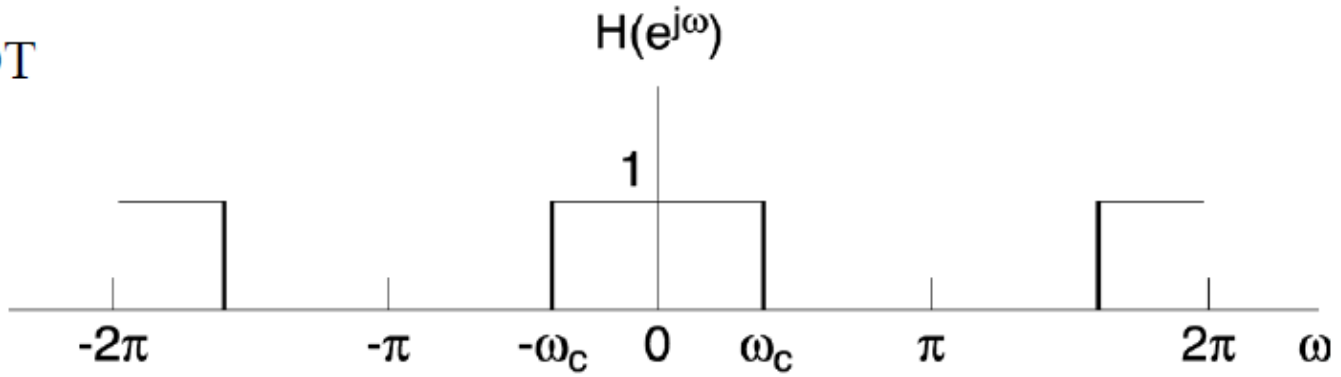
## Idealized Filters

### Lowpass filter

CT




DT

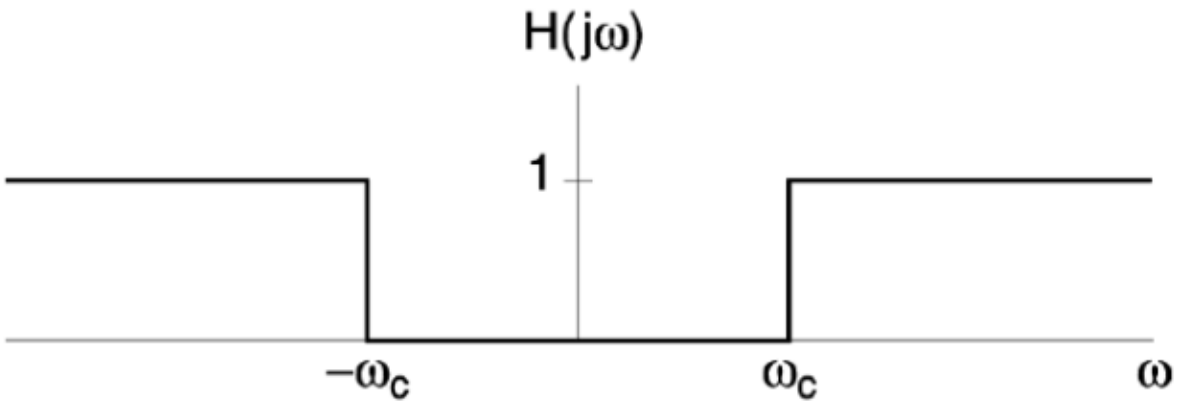



Note: $|H| = 1$ and $\angle H = 0$ for the ideal filters in the passbands, no need for the phase plot.

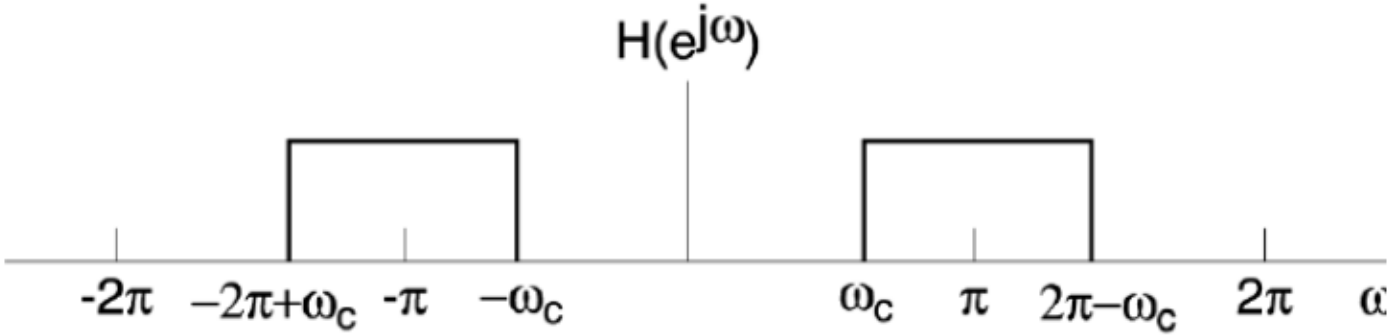
# Filtering

**Highpass**

CT



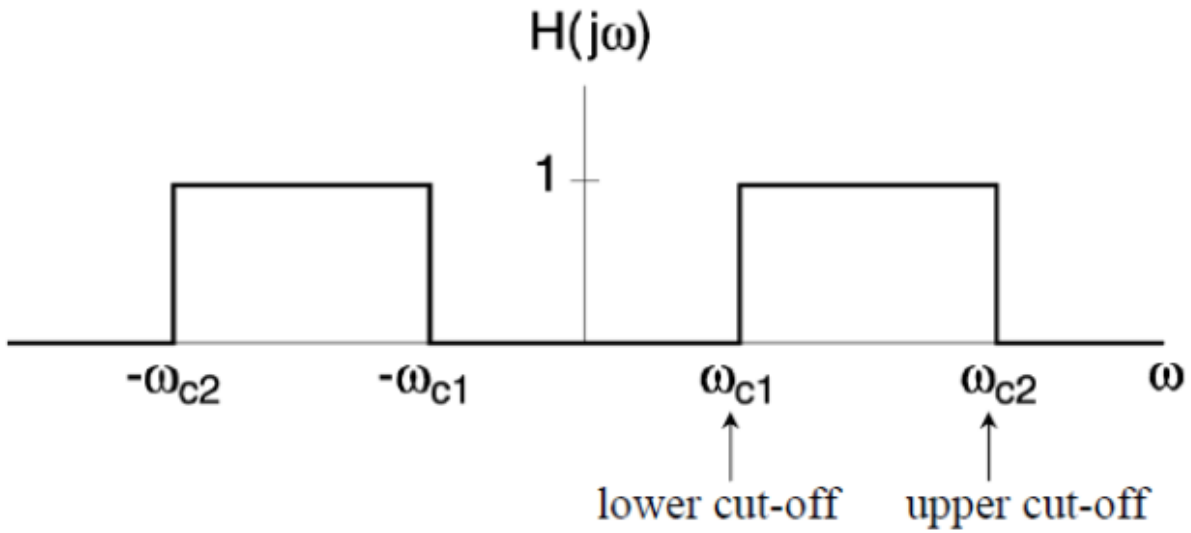

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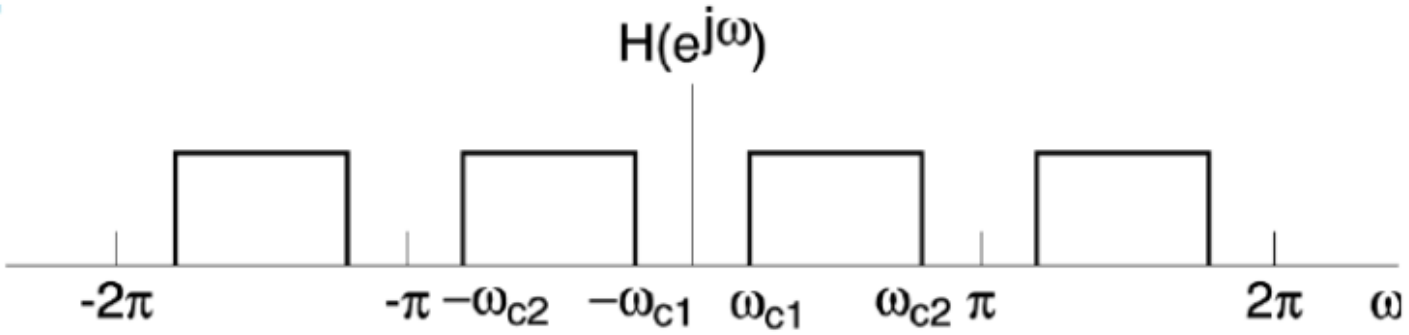
# Filtering

## Bandpass

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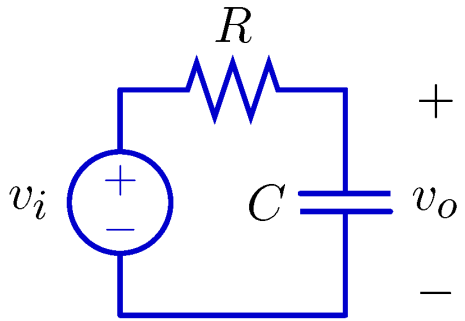



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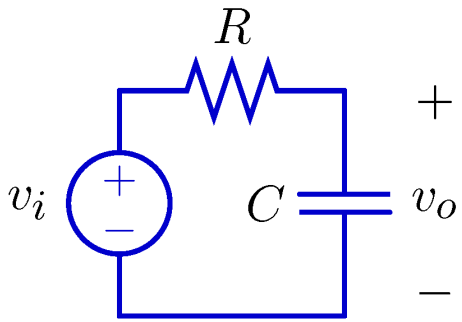
# Filtering

Example: Low-Pass Filtering with an RC circuit

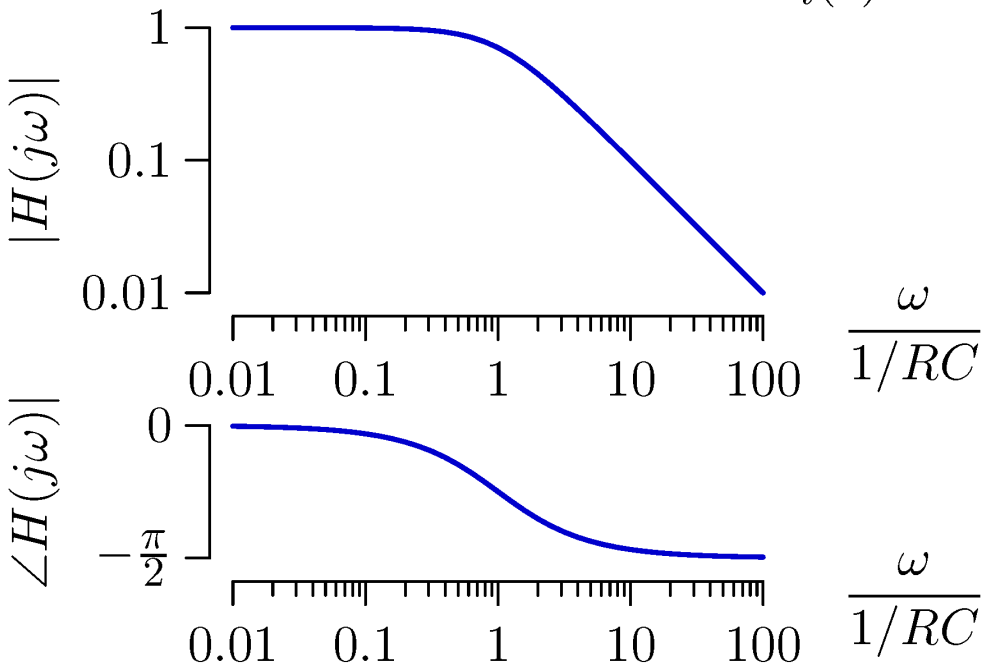


# Lowpass Filter

Calculate the frequency response of an RC circuit.

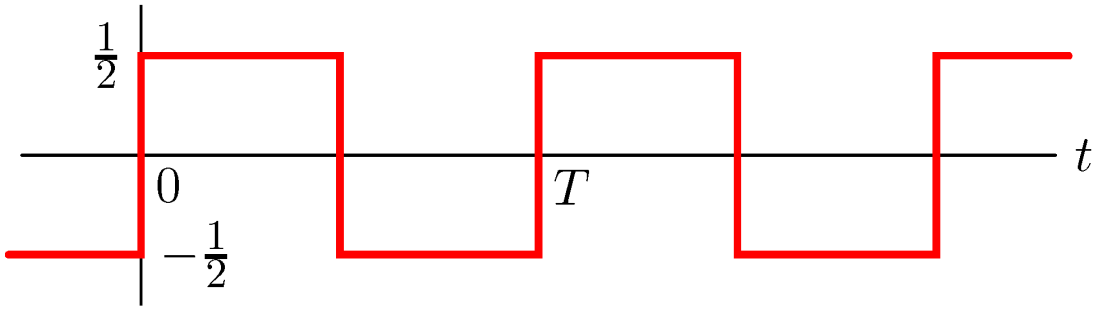



$$\begin{aligned}
\text{KVL:} \quad & v_i(t) &&= Ri(t) + v_o(t) \\
\text{C:} \quad & i(t) &&= C\dot{v}_o(t) \\
\text{Solving:} \quad & v_i(t) &&= RC\dot{v}_o(t) + v_o(t) \\
& V_i(s) &&= (1 + sRC)V_o(s) \\
& H(s) &&= \frac{V_o(s)}{V_i(s)} = \frac{1}{1 + sRC}
\end{aligned}$$

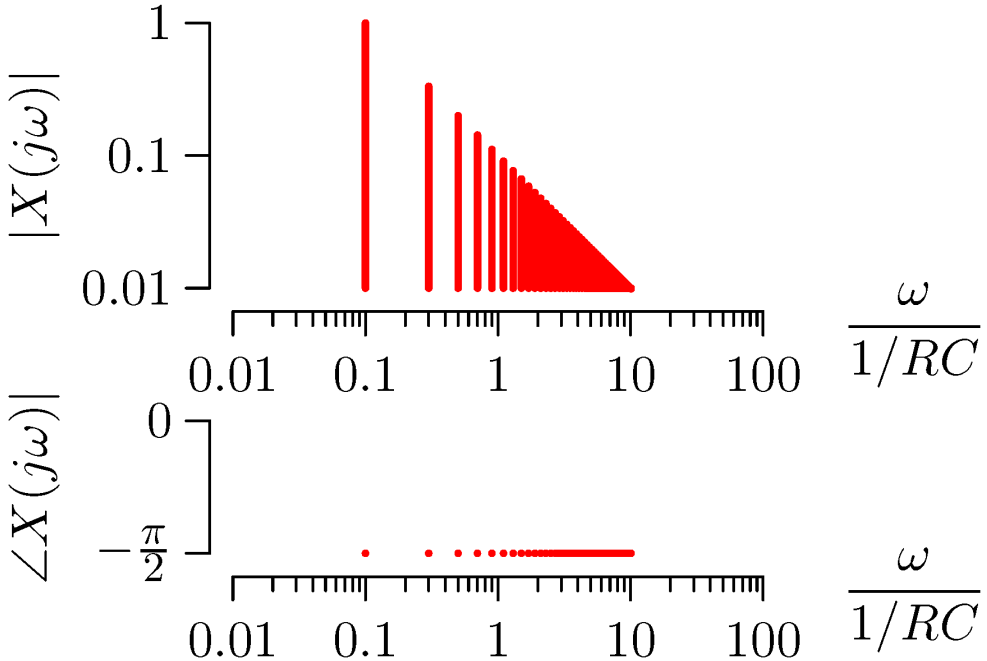


# Lowpass Filtering

Let the input be a square wave.

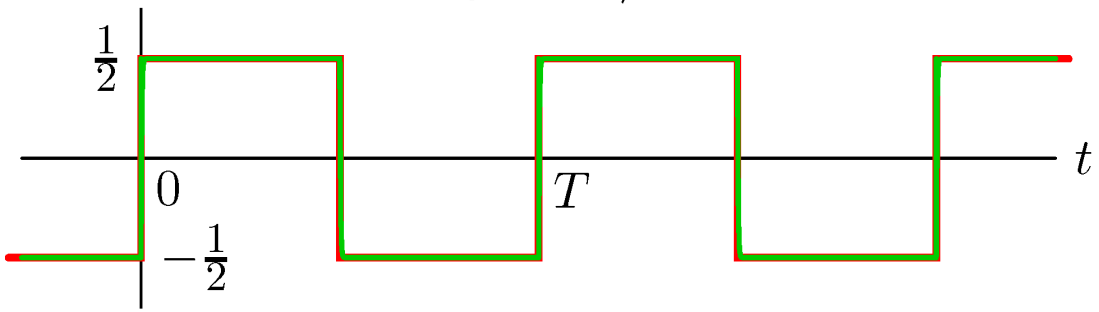



$$x(t) = \sum_{k \text{ odd}} \frac{1}{j\pi k} e^{j\omega_0 kt} \; ; \quad \omega_0 = \frac{2\pi}{T}$$

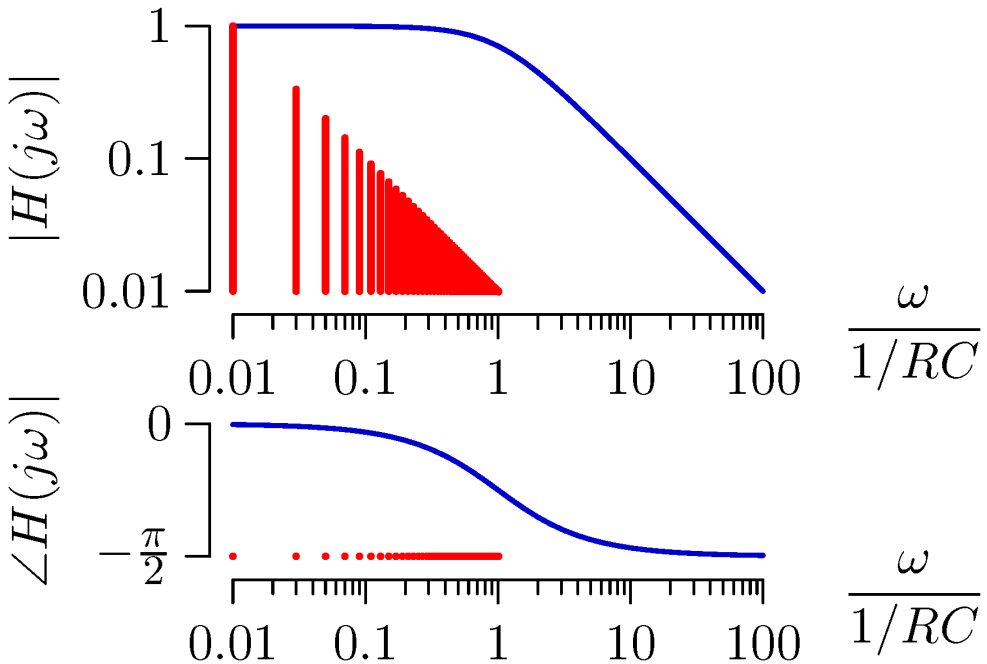


# Lowpass Filtering

Low frequency square wave: $\omega_0 << 1/RC$.

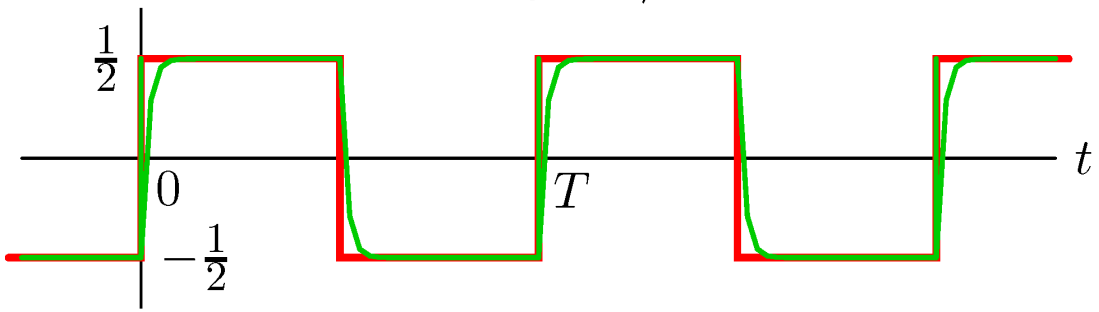



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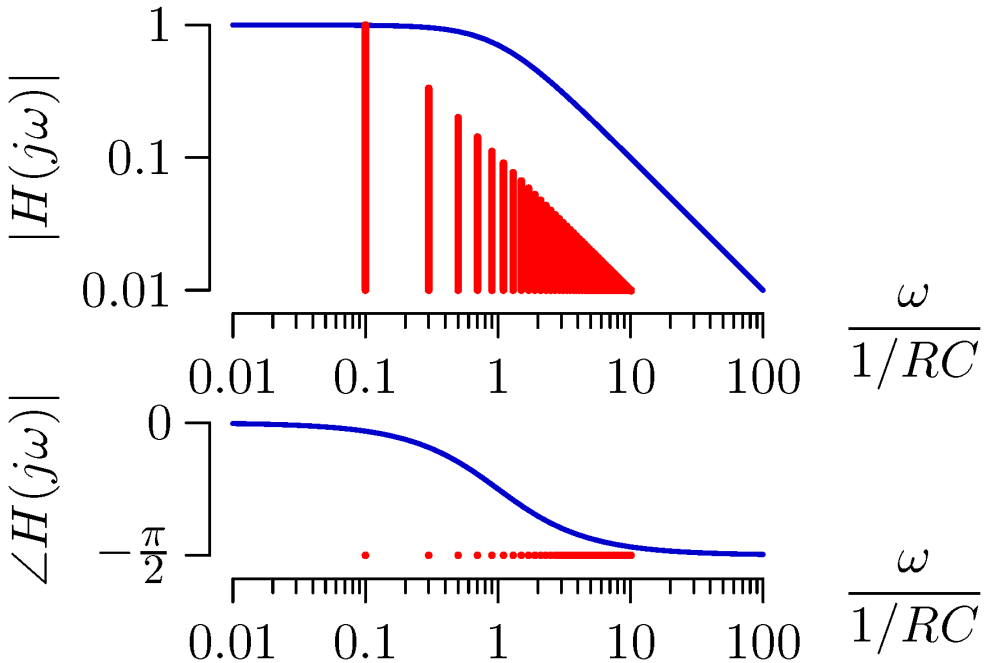


# Lowpass Filtering

Higher frequency square wave: $\omega_0 < 1/RC$.

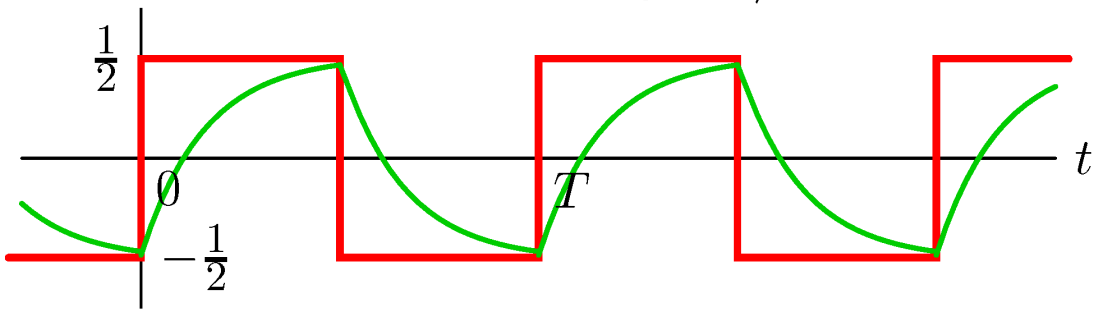



$$x(t) = \sum_{k \text{ odd}} \frac{1}{j\pi k} e^{j\omega_0 kt} \; ; \quad \omega_0 = \frac{2\pi}{T}$$

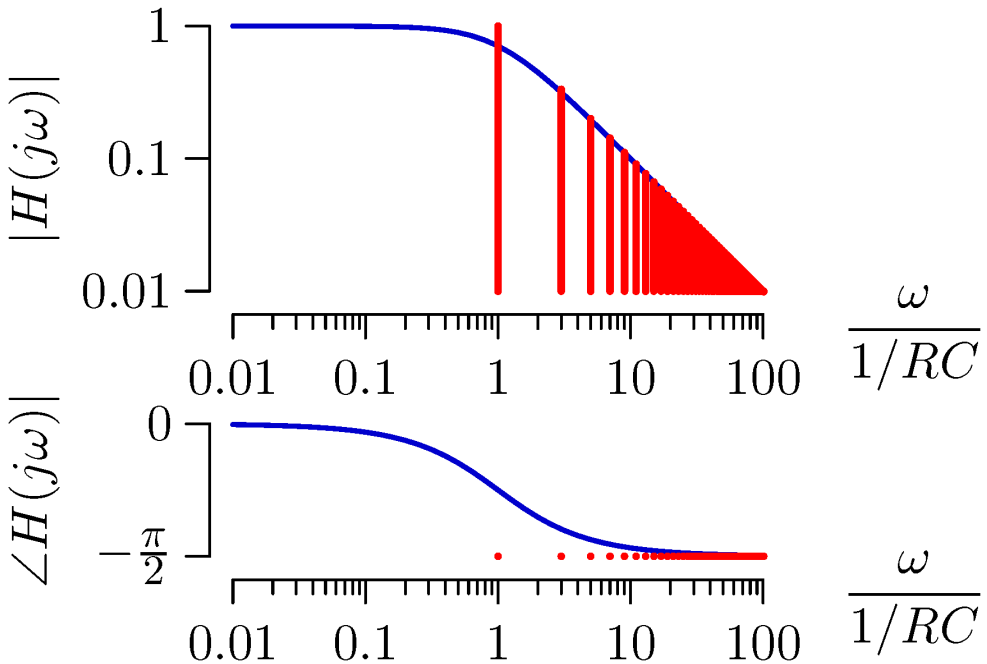


# Lowpass Filtering

Still higher frequency square wave: $\omega_0 = 1/RC$.

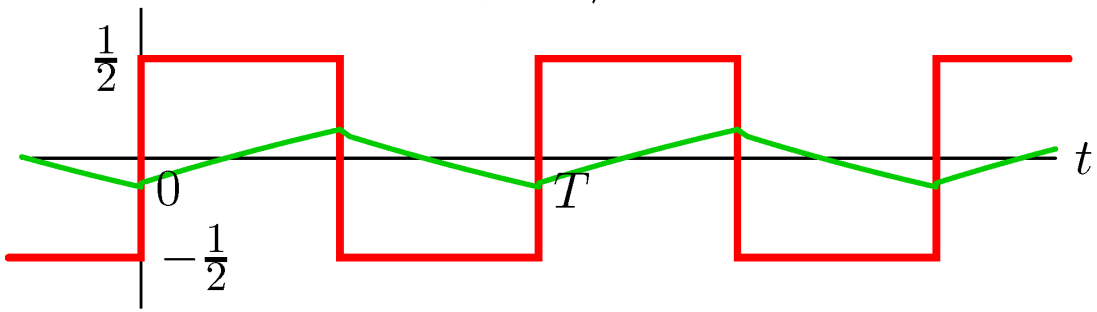



$$x(t) = \sum_{k \text{ odd}} \frac{1}{j\pi k} e^{j\omega_0 k t} \quad ; \quad \omega_0 = \frac{2\pi}{T}$$

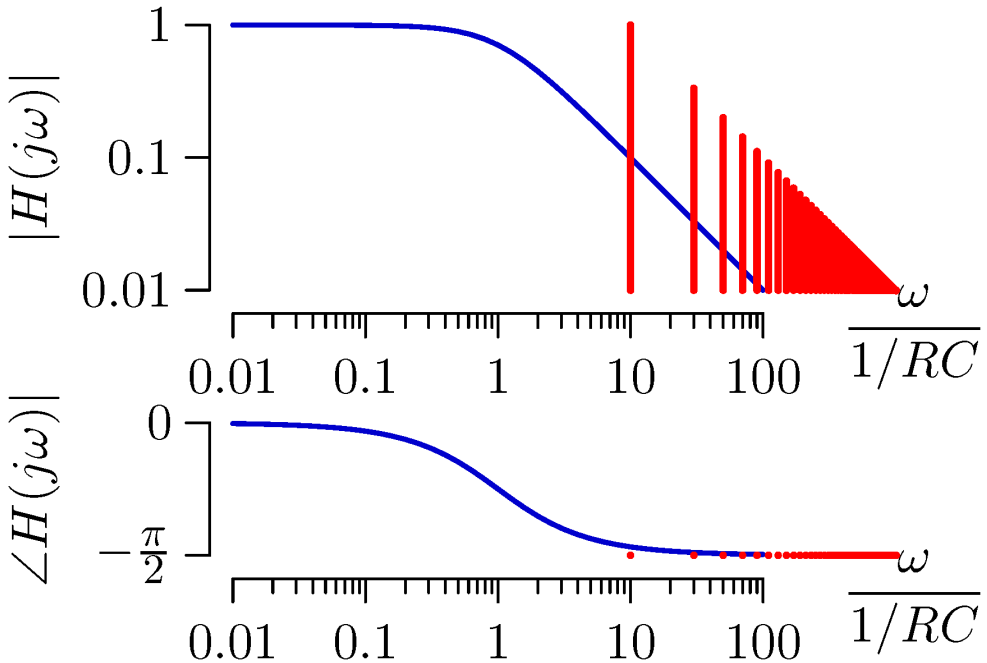


# Lowpass Filtering

High frequency square wave: $\omega_0 > 1/RC$.




$$x(t) = \sum_{k \text{ odd}} \frac{1}{j\pi k} e^{j\omega_0 k t} \quad ; \quad \omega_0 = \frac{2\pi}{T}$$



# Properties of CTFS

- Linearity, time reversal, symmetry, time shift, time derivative → making it easier to find the Fourier coefficients of a new signal
- The Complex Exponential Form is much easier in math
- Get more familiar with the CE form

# Properties of CTFS: Linearity

Consider $y(t) = Ax_1(t) + Bx_2(t)$, where $x_1(t)$ and $x_2(t)$ are periodic in $T$. What are the CTFS coefficients $c_k$, in terms of $a_k$ and $b_k$?

**Q: Is $y(t)$ periodic in $T$?**

$$
\begin{aligned}
c_k &= \frac{1}{T}\int_T y(t)e^{-j\frac{2\pi kt}{T}}\,dt \\
&= \frac{1}{T}\int_T \left(Ax_1(t) + Bx_2(t)\right)e^{-j\frac{2\pi kt}{T}}\,dt \\
&= A\cdot\frac{1}{T}\int_T x_1(t)e^{-j\frac{2\pi kt}{T}}\,dt + B\cdot\frac{1}{T}\int_T x_2(t)e^{-j\frac{2\pi kt}{T}}\,dt \\
&= Aa_k + Bb_k
\end{aligned}
$$

If $y(t) = Ax_1(t) + Bx_2(t)$, then $c_k = Aa_k + Bb_k$

# Properties of CTFS: Time flip (reversal)

Consider $y(t) = x(-t)$, where $x(t)$ is periodic in $T$. What are the CTFS coefficients $b_k$, in terms of $a_k$? First, $y(t)$ must also be periodic in $T$.

$$x(t) = \sum_{k=-\infty}^{\infty} a_k e^{j\frac{2\pi kt}{T}}$$

$$y(t) = x(-t) = \sum_{k=-\infty}^{\infty} a_k e^{j\frac{2\pi k(-t)}{T}} = \sum_{k=-\infty}^{\infty} a_k e^{-j\frac{2\pi kt}{T}}$$

# Properties of CTFS: Time flip (reversal)

Let $m = -k$, then:

$$y(t) = x(-t) = \sum_{m=-\infty}^{\infty} a_{-m} e^{j\frac{2\pi mt}{T}}$$

Since we know:

$$y(t) = \sum_{m=-\infty}^{\infty} b_m e^{j\frac{2\pi mt}{T}}$$

We have the following equivalence:

$$b_k = a_{-k}$$

**Q: What happens when the signal is real?**

If $y(t) = x(-t)$, then:

$$b_k = a_{-k} = a_k^*$$

# Properties of CTFS: Symmetric and Antisymmetric Parts

Any arbitrary signal $x(t)$ can be written into two parts:

$$x(t) = x_o(t) + x_e(t)$$

- $x_e(t)$ is **symmetric** about $t = 0$, if $x_e(t) = x_e(-t)$ for all $t$.
- $x_o(t)$ is **antisymmetric** about $t = 0$, if $x_o(t) = -x_o(-t)$ for all $t$.

The symmetric and antisymmetric components are given by:

$$x_e(t) = \frac{x(t) + x(-t)}{2}, \quad x_o(t) = \frac{x(t) - x(-t)}{2}$$

If $x(t) = x_e(t) + x_o(t)$ is a real-valued signal and periodic in time with fundamental period $T$, what are the Fourier Series coefficients of $x_e(\cdot)$ and $x_o(\cdot)$, in terms of $a_k$?

# Properties of CTFS: Symmetric and Antisymmetric Parts

If $x(t) = x_e(t) + x_o(t)$ is a real-valued signal and periodic in time with fundamental period $T$, what are the Fourier coefficients of $x_e(\cdot)$ and $x_o(\cdot)$, in terms of $a_k$?

**If $x(t)$ is real valued periodic signal, $a_k = a^*_{-k}$**

$$x_e(t) = \frac{x(t) + x(-t)}{2} \quad \text{Linearity and time flip} \quad a^e_k = \frac{a_k + a_{-k}}{2}$$

$$x_o(t) = \frac{x(t) - x(-t)}{2} \quad \text{Linearity and time flip} \quad a^o_k = \frac{a_k - a_{-k}}{2}$$

$$a_k^e = \frac{a_k + a_k^*}{2} = \mathrm{Re}(a_k)$$

$$a_k^o = \frac{a_k - a_k*}{2} = j\mathrm{Im}(a_k)$$

The real part of $a_k$ comes from the symmetric part of the signal, the imaginary part of $a_k$ comes from the antisymmetric part of the signal.

# Properties of CTFS: Time Shift

Consider $y(t) = x(t - t_0)$, where $x(t)$ is periodic in $T$. What are the CTFS coefficients $b_k$, in terms of $a_k$?

$$b_k = \frac{1}{T}\int_T y(t)e^{-j\frac{2\pi kt}{T}}dt = \frac{1}{T}\int_T x(t-t_0)e^{-j\frac{2\pi kt}{T}}dt$$

Let $u = t - t_0$, then $t = u + t_0$, and $dt = du$, so the integral becomes:

$$= \frac{1}{T}\int_T x(u)e^{-j\frac{2\pi k(u+t_0)}{T}}du$$

$$= \frac{1}{T}\int_T x(u)e^{-j\frac{2\pi ku}{T}}e^{-j\frac{2\pi kt_0}{T}}du$$

Since $e^{-j\frac{2\pi kt_0}{T}}$ is a constant, we get:

$$= e^{-j\frac{2\pi kt_0}{T}} \cdot \frac{1}{T} \int_T x(u) e^{-j\frac{2\pi ku}{T}} du$$

$$= e^{-j\frac{2\pi kt_0}{T}} a_k$$

Each coefficient $b_k$ in the series for $y(t)$ is a constant $e^{-jk\omega_0 t_0}$ times the corresponding coefficient $a_k$ in the series for $x(t)$.

# Properties of CTFS: Time Derivative

Consider $y(t) = \frac{d}{dt}x(t)$, where $x(t)$ and $y(t)$ are periodic in $T$. What are the CTFS coefficients $b_k$, in terms of $a_k$? Start with the synthesis equation:

$$x(t) = \sum_{k=-\infty}^{\infty} a_k e^{j\frac{2\pi kt}{T}}$$

Then, from the definition of $y(t)$, we have:

$$y(t) = \dot{x}(t) = \frac{d}{dt}\left(\sum_{k=-\infty}^{\infty} a_k e^{j\frac{2\pi kt}{T}}\right) = \sum_{k=-\infty}^{\infty}\left(j\frac{2\pi k}{T}a_k\right)e^{j\frac{2\pi kt}{T}}$$

$$= \sum_{k=-\infty}^{\infty} b_k e^{j\frac{2\pi kt}{T}}$$

From this form, we can see that:

$$b_k = j\frac{2\pi k}{T}a_k$$

# Properties of CTFS: Summary

- **Linearity**: If $y(t) = Ax_1(t) + Bx_2(t)$, then

$$c_k = Aa_k + Bb_k$$

- **Time reversal**: If $y(t) = x(-t)$, then

$$b_k = a_{-k}$$

- **Time shift**: If $y(t) = x(t - t_0)$, then

$$b_k = e^{-j\frac{2\pi k t_0}{T}} a_k$$

- **Time derivative**: If $y(t) = \dot{x}(t)$, then

$$b_k = j\frac{2\pi k}{T} a_k$$

# Fourier Series: Summary

Fourier series represent signals by their frequency content.

Representing a signal by its frequency content is useful for many signals, e.g., music.

Fourier series motivate a new representation of a system as a filter.